

DNA / Protein Sequence Analysis Using BLAST

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Domain: Bioinformatics

Task: Task 1 – DNA / Protein Sequence Analysis

Tool Used: NCBI BLASTp

1. Introduction

Bioinformatics serves as a scientific discipline that merges biology with computer science and information technology to study biological data. The primary purpose of bioinformatics protein sequence analysis enables scientists to study protein function and structure together with protein evolutionary connections. The National Center for Biotechnology Information (NCBI) created the Basic Local Alignment Search Tool (BLAST) which serves as a popular bioinformatics tool. The tool enables users to compare biological sequences with extensive databases in order to find matching or similar sequences. The research utilized BLASTp to assess a protein sequence and discover corresponding proteins throughout various species.

2. Objective

The objectives of this task are: To retrieve a protein sequence from the NCBI database To perform BLASTp analysis To identify homologous protein sequences To analyze similarity parameters such as percent identity, query coverage, E-value, and alignment score

3. Materials and Tools Used

NCBI Protein Database

NCBI BLAST (BLASTp)

Web Browser MS Word / Google Docs

4. Methodology

4.1. Protein Sequence Selection

The protein hemoglobin subunit alpha [Homo sapiens] was selected for analysis. Hemoglobin serves as a crucial protein that transports oxygen throughout the human body while its molecular structure remains unchanged across various species. The reference sequence used was: NCBI Reference Sequence: NP_000508.1 Sequence

Length: 142 amino acids.

NIH National Library of Medicine
National Center for Biotechnology Information

Protein

Genescript

hemoglobin subunit alpha [Homo sapiens]

NCBI Reference Sequence: NP_000508.1

[Identical Proteins](#) [FASTA](#) [Graphics](#)

Go to:

NP_000508 142 aa 11aaaa PRI 26-NOV-2025

DEFINITION hemoglobin subunit alpha [Homo sapiens].

ACCESSION NP_000508

VERSION NP_000508.1

KEYWORDS RefSeq; NM_000507.6

SOURCE Homo sapiens (human)

ORGANISM Homo sapiens

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorhina; Catarrhini; Hominoidea; Homo.

REFERENCE 1. (residues 1 to 142)

AUTHORS Schaffer, J.V., Ma, M., Qian, G., Ma, K.M., Pei, A., Sui, B., Latham, A.P., Portano Valdes, L., Tai, D., Matsumoto, M., Li, X., Bachelder, R., Cesnik, A., Gao, J., Li, F., Kellomaki, M., Pei, A., Pajdlo, E., Qin, Y., Zhou, X., Chaves, C., Lenikiewicz, J., Chen, J., Qiu, K., Pratt, D., Zapp, P., Schenck, J., Sali, A., Harper, J.W., Ogilvie, P., Foster, L.J., Huttlain, L., Lundberg, L. and Idkay, I.

TITLE Multimeric cell map as a foundation for structural and functional genomics

Change region shown

Customize view

Analyze this sequence

Run BLAST

Identify Conserved Domains

Highlight Sequence Features

Find in this Sequence

Show in Genome Data Viewer

Protein 3D Structure

3D structure of human hemoglobin in the R conformation

PDB: 5VIC

Source: Homo sapiens

Method: Electron Microscopy

Resolution: 2.45 Å

See all 320 structures

4.2. FASTA Sequence Retrieval

The NCBI Protein database provided the protein sequence which researchers retrieved in FASTA format. The FASTA format begins with a header line which is followed by an amino acid sequence that BLAST analysis can utilize.

NIH National Library of Medicine
National Center for Biotechnology Information

Protein

FASTA

hemoglobin subunit alpha [Homo sapiens]

NCBI Reference Sequence: NP_000508.1

[GenPept](#) [Identical Proteins](#) [Graphics](#)

>NP_000508.1 hemoglobin subunit alpha [Homo sapiens]
NVLSPOKTYKAWGKYGAGEYGAELRNFLSTPTKTYTPHDLHGAGKQWGDQKVDALNIA
VAWDQRMWALSLSQHWRLRYGPNFKLSKCLLVTLAWLPREFTPWMSLQKFLASVSTVLTSA
YR

Change region shown

Analyze this sequence

Run BLAST

Identify Conserved Domains

Show in Genome Data Viewer

Protein 3D Structure

3D structure of human hemoglobin in the R conformation

PDB: 5VIC

Source: Homo sapiens

Method: Electron Microscopy

Resolution: 2.45 Å

See all 320 structures

Articles about the HBA2 gene

Interactions of electrophoretically silent hemoglobin Hekinan II [HBM] [J. Ryan J. Med. 2024]

4.3. BLASTp Analysis

The researchers used BLASTp to examine the protein sequence against the NCBI non-redundant protein database with default parameters. The researchers pasted the FASTA sequence into the BLASTp query box to execute their analysis.

BLAST[®] » blastp suite

Standard Protein BLAST

Enter Query Sequence

Enter accession number(s), gis, or FASTA sequence(s) [Clear](#)

Query subrange

From: To:

Or, upload file

Choose file | No file chosen

Job Title

Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database: ClusteredNR (nr_cluster_seq)

Organism: [Add organism](#)

Exclude: ☐ Modes (MMXP) ☐ Non-redundant RefSeq proteins (WPI) ☐ Uncultured/environmental sample sequences

Entrez Query

Enter an Entrez query to limit search [Create custom database](#)

Program Selection

Algorithm: ☐ Quick BLAST (Accelerated protein-protein BLAST) ☒ blastp (protein-protein BLAST)

5. Results

The BLASTp analysis produced multiple significant matches which displayed high similarity to hemoglobin proteins found in various organisms.

Clusters producing significant alignments

Download Select columns Show 100

☒ select all 100 clusters selected

GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒ 6 member(s), 1 organism(s)	human	Chain A, HEMOGLOBIN-BASED BLOOD SUBSTITUTE (Homo...	284	567	99%	2e-94	100.00%	283	1ABW_A
☒ 1 member(s), 1 organism(s)	Francosa's langur	translation initiation factor (IF-2-like) [Taxhydroceus francos...	284	284	100%	6e-94	97.18%	327	XP_033092340.1
☒ 9 member(s), 9 organism(s)	primates	TPA, globin C1 (Macaca mulatta)	276	276	100%	9e-93	96.48%	177	SAI82172.1
☒ 1 member(s), 1 organism(s)	cotton-top tamarin	hypothetical protein P7K49_035633 (Saquinus oedipus)	274	274	100%	6e-92	94.37%	193	KAK2008208.1
☒ 1 member(s), 1 organism(s)	slow loris	hemoglobin subunit alpha-like (Nycticebus coucang)	270	270	100%	3e-90	93.66%	177	XP_053412047.1
☒ 24 member(s), 17 organism(s)	cellular organisms	MULTISPECIES hemoglobin alpha subunit family protein, part...	268	268	100%	5e-90	90.85%	150	WP_105150429.1
☒ 1 member(s), 1 organism(s)	human	Hemoglobin subunit alpha (Homo sapiens)	265	265	100%	3e-88	75.94%	187	AQN67053.1
☒ 124 member(s), 57 organism(s)	placental	Hemoglobin subunit alpha (Paoia anubis)	262	262	100%	1e-87	92.25%	142	NP_001102287.1
☒ 11 member(s), 8 organism(s)	bats	hemoglobin subunit alpha (Myotis yumanensis)	259	259	100%	7e-87	86.73%	142	XP_070269501.1
☒ 26 member(s), 11 organism(s)	Rodents and rabbits	hemoglobin alpha chain (Alexandromys oeconomus)	258	258	100%	3e-86	89.44%	142	AFX00018.1
☒ 1 member(s), 1 organism(s)	hamadryas baboon	TPA, globin C1 (Papio hamadryas)	264	264	100%	8e-86	92.25%	327	SAI82176.1
☒ 4 member(s), 4 organism(s)	bats	hemoglobin subunit alpha (Rhinolophus ferrumequinum)	256	256	100%	1e-85	87.32%	142	XP_032950812.1
☒ 141 member(s), 35 organism(s)	rodents	hemoglobin alpha adult chain 2 (Mus musculus)	255	255	100%	5e-85	86.62%	142	NP_001077424.1
☒ 10 member(s), 6 organism(s)	primates	RecName: Full=Hemoglobin subunit alpha-2; AltName: Full=Al...	254	254	99%	6e-85	87.94%	141	P20019.1
☒ 1 member(s), 1 organism(s)	bats	PREDICTED hemoglobin subunit alpha (Minioctopus natalkensis)	254	254	100%	1e-84	86.62%	142	XP_016071731.1
☒ 56 member(s), 32 organism(s)	even-toed ungulates & wh...	hemoglobin subunit alpha (Bos taurus)	252	252	100%	8e-84	88.03%	142	NP_001070890.2
☒ 1 member(s), 1 organism(s)	odd-toed ungulates	RecName: Full=Hemoglobin subunit alpha; AltName: Full=Alph...	251	251	99%	1e-83	87.23%	141	P09006.1
☒ 1 member(s), 1 organism(s)	eastern chipmunk	RecName: Full=Hemoglobin subunit alpha (Tamias striatus)	251	251	99%	2e-83	87.23%	141	R3FW19.1

5.1. BLAST Result Parameters Percent Identity:

Measures exact amino acid matches Query Coverage: Indicates how much of the query sequence is aligned E-value: Represents statistical significance (lower is better) Alignment Score: Reflects alignment quality.

Protein Name	Organism	Percent Identity	Query Coverage	E-value
Hemoglobin subunit alpha	Homo sapiens	100%	99%	2e-94
Hemoglobin subunit alpha-like	Trachypithecus francoisi	97.18%	100%	6e-94
Globin C1	Macaca mulatta	96.48%	100%	9e-93
Hypothetical protein	Saguinus oedipus	94.37%	100%	8e-92
Hemoglobin subunit alpha-like	Nycticebus coucang	93.66%	100%	3e-90

Alignment Analysis

The alignment results show that amino acid sequences between different species maintain strong conservation. The E-values show very low values which prove the similarities between the two entities exist as statistically significant results that don't happen by random chance. The conserved regions of hemoglobin show which parts of the protein perform vital biological functions.

Discussion

The BLASTp results demonstrate that hemoglobin subunit alpha shows high conservation across multiple species with particular emphasis on primates and mammals. The high percent identity combined with complete query coverage indicates strong evolutionary connections between the two species. The conserved regions of the protein function as essential components for oxygen binding and transport, which demonstrates the vital role of hemoglobin in all living organisms.

Conclusion

The research work used BLASTp because it provides effective protein sequence analysis tools. The hemoglobin subunit alpha protein showed high similarity with homologous proteins from multiple species. The research demonstrates how bioinformatics tools such as BLAST enable scientists to study protein evolution through their conserved functions.

TASK 2: Multiple Sequence Alignment (MSA)

Tool Used: Clustal Omega (EMBL-EBI)

Introduction

Bioinformatics uses Multiple Sequence Alignment (MSA) as a basic method that enables researchers to compare three or more biological sequences to find matching sequence segments. The detected similarities between sequences show their shared functional and structural and evolutionary ties. Researchers use MSA to examine conserved parts of biological systems and to identify crucial functional elements and to study how species maintain their evolutionary patterns. Researchers have extensively studied hemoglobin as the protein that carries oxygen through blood. Scientists use the hemoglobin alpha subunit because its sequence remains unchanged across all vertebrate species to study multiple sequence alignment. The researchers used Clustal Omega to align hemoglobin alpha protein sequences from five different species which allowed them to find shared amino acid sequences while they studied their functional roles.

objectives

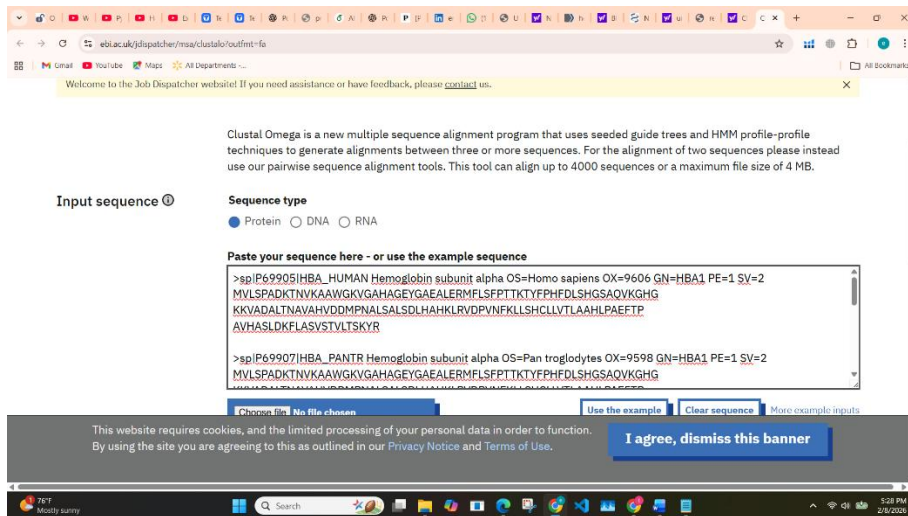
The objectives of this task are: to select protein sequences which exist in the same protein family to use Clustal Omega for Multiple Sequence Alignment to identify amino acid regions which show conservation across different species to explain the biological functions of regions which show conservation in their structure

Materials and Tools Used

- ☐ **UniProt Database** – for protein sequence retrieval
- ☐ **Clustal Omega (EMBL-EBI)** – for multiple sequence alignment
- ☐ **Web Browser**

Methodology

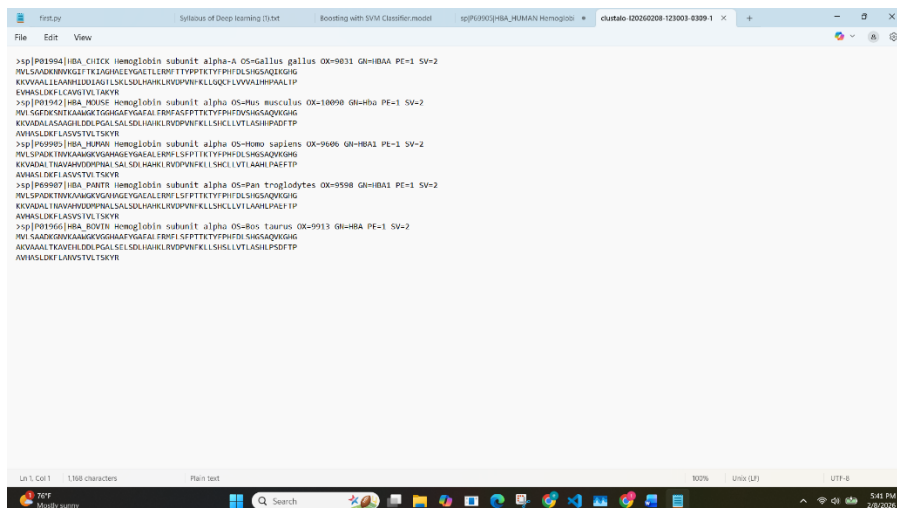
The **hemoglobin alpha protein family** was selected for this study because all members perform the same biological function of oxygen binding and transport. Selecting proteins from the same family but different species helps in understanding evolutionary conservation.



Protein Sequences Used

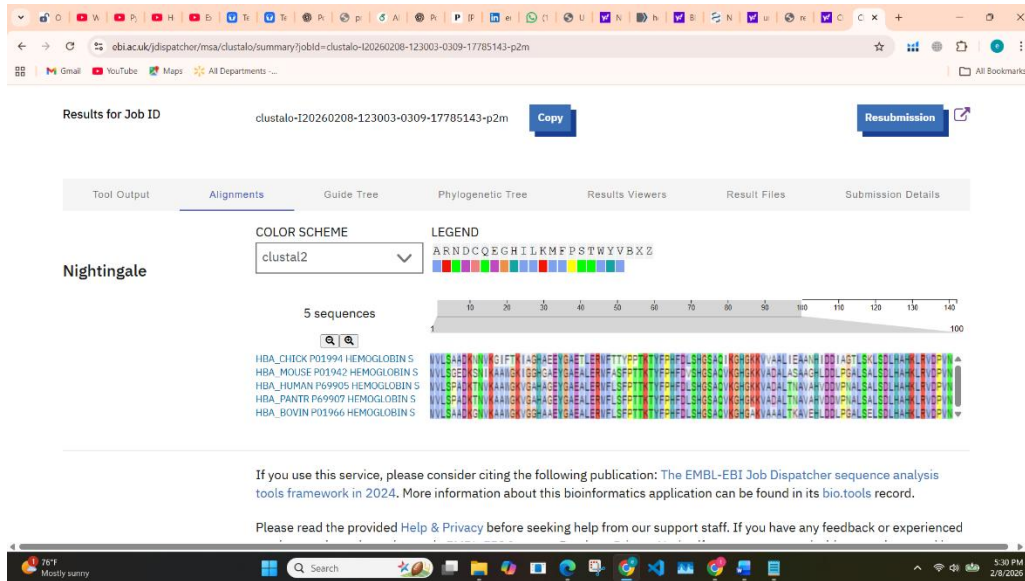
Five hemoglobin alpha protein sequences were selected from UniProt in FASTA format:

Protein	Species
Hemoglobin subunit alpha	<i>Homo sapiens</i>
Hemoglobin subunit alpha	<i>Pan troglodytes</i>
Hemoglobin subunit alpha	<i>Mus musculus</i>
Hemoglobin subunit alpha	<i>Bos taurus</i>
Hemoglobin subunit alpha	<i>Gallus gallus</i>



Multiple Sequence Alignment

The collected FASTA sequences were submitted to **Clustal Omega**, an online multiple sequence alignment tool provided by EMBL-EBI. Default parameters were used for alignment. The output was visualized using the colored alignment (Nightingale view), which highlights conserved and variable regions using color coding.



Results

The multiple sequence alignment successfully aligned all five hemoglobin alpha sequences. The alignment showed a high degree of similarity across most regions of the sequences.

Conserved Regions

Several amino acid residues were conserved across all five sequences. These conserved regions were identified by:

- Identical amino acids appearing in the same positions
- Similar color patterns across aligned sequences

The high conservation indicates that these amino acids play an essential role in maintaining protein structure and function.



TASK 4: Mini Project – Bioinformatics Study

Overview of Bioinformatics Databases

National Center for Biotechnology Information NCBI serves as a free bioinformatics platform which the National Institutes of Health NIH in the USA operates. The system provides users with direct access to multiple biological databases which contain information about DNA and RNA and protein sequences. The NCBI database system includes essential databases which include GenBank and PubMed and Protein databases and Gene databases and BLAST. GenBank is recognized as one of the most extensive nucleotide sequence databases which exist globally. The NCBI platform offers BLAST as a powerful tool which enables users to conduct sequence similarity searches.

NCBI Applications include: Users can retrieve DNA and protein sequences from the system. BLAST enables users to perform sequence similarity analysis. Users can conduct literature searches through the PubMed system. Researchers use the system for gene annotation and genomic research.

UniProt (Universal Protein Resource) functions as a complete database which contains information about protein sequences together with their functional characteristics. The project between multiple bioinformatics institutes creates a database which contains high-quality protein information that researchers can access. The main components of UniProt include two primary sections: Swiss-Prot: The section contains protein entries which have undergone manual review and direct curation. TrEMBL: The section contains protein entries which have not undergone manual review and yet have received automated protein annotations. UniProt supplies extensive details concerning protein functions and structural aspects and protein domain information and post-translational modification data.

Applications of UniProt: Researchers use it to analyze protein sequences. Scientists use it to identify protein functions. Researchers study protein families and protein domains through this method. Scientists use this resource as their primary reference material for proteomics studies.

PDB (Protein Data Bank) The Protein Data Bank (PDB) serves as a dedicated database which maintains three-dimensional structural data for biological macromolecules that include proteins and nucleic acids. Scientists obtain these structures through various experimental methods which include X-ray crystallography and NMR spectroscopy and cryo-electron microscopy. The research community uses PDB for both structural biology and drug discovery research purposes. Scientists can use it to view molecular structures and perform detailed molecular structure analysis.

Applications of PDB: Scientists use PDB to visualize protein structures Researchers use PDB in their efforts to design drugs which depend on specific structural information Scientists use PDB to study how proteins and small molecules interact with each other Researchers use PDB for their studies of both protein structure and protein function.

Feature	NCBI	UniProt	PDB
Data Type	DNA, RNA, Protein	Protein	3D Structures
Main Focus	Genomic & literature data	Protein function	Structural biology
Curation	Automatic & manual	Highly curated	Experiment-based
Tools Provided	BLAST, PubMed	Protein annotation tools	Structure visualization
Usage	Sequence analysis	Proteomics research	Drug discovery

The section on **advantages** presents evidence for all free public databases which academic researchers use throughout the world as their primary research resource which provides them with reliable biological data and enables them to conduct multiple bioinformatics test procedures.

The section describes two main **drawbacks** of the system, which include its difficulty for new users to handle its massive data volume and its requirement that users possess initial bioinformatics skills. The system depends on automated database systems which

produce annotations, though these annotations contain potential errors. The system needs to undergo continuous updates because its current state requires such maintenance to reach correct system operation.

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Future Scope

The future of bioinformatics databases will depend on their ability to incorporate cutting-edge technologies that include artificial intelligence and machine learning. The performance of data annotation and prediction systems and data analysis operations will receive improvements through these technologies. The upcoming events will exhibit the following developments: AI-based automatic data curation Integration of multi-omics data Personalized medicine applications Improved visualization and user-friendly interfaces The upcoming developments will increase the value of bioinformatics databases for research purposes and healthcare applications.

The research conclusion The mini project compared three major bioinformatics databases which include NCBI and UniProt and PDB. The three databases contribute distinct fundamental functions which researchers need to conduct their biological studies. NCBI provides comprehensive genomic and literature data, UniProt focuses on protein sequence and functional information, and PDB specializes in three-dimensional structural data. The databases provide support for multiple bioinformatics applications which include drug discovery and sequence analysis. Researchers should study the databases' features and advantages and limitations in order to select the most suitable database for their research requirements.

Role of Bioinformatics in Drug Discovery

Abstract

The drug discovery process requires extensive laboratory testing because it takes a long time and costs a lot of money to complete. The development of computational sciences has created bioinformatics as a strong tool which speeds up and enhances the drug discovery process. Bioinformatics uses biological information together with computer science and data analysis techniques to handle and analyze extensive biological data collections. The mini project demonstrates how bioinformatics supports every phase of drug discovery, which includes target identification, target validation, lead discovery, drug design, and optimization. The study presents the benefits and drawbacks of bioinformatics-based drug discovery together with its future development possibilities. Bioinformatics knowledge enables researchers to create drugs which work better and develop more rapidly and have fewer safety issues.

Introduction

The process of drug discovery aims to find new medications which can treat various medical conditions. The traditional method of drug discovery relies on a trial-and-error approach to experiments that requires significant financial investment and can take several years before achieving successful results. The increasing volume of biological data which results from genome sequencing and proteomics and molecular research now requires researchers to use computational methods for their analyses. Modern drug discovery relies on bioinformatics to process biological information through its computational systems and database resources. The system enables researchers to investigate disease processes at the molecular level while discovering new drug targets and developing new drug compounds. Bioinformatics provides three benefits to drug development which include cost reduction and time efficiency and higher success rates of drug development projects.

Role of Bioinformatics in Drug Discovery

3.1 Target Identification

Target identification represents the essential initial step for drug discovery. A drug target is usually a protein, gene, or enzyme associated with a disease. Bioinformatics aids researchers through its ability to: Analyze genomic and proteomic data Determine which genes cause diseases Examine metabolic and signaling pathways Study the differences between healthy cells and diseased cells Researchers use NCBI and UniProt databases to discover and investigate potential drug targets.

Target Validation The process begins with target identification which requires target validation to establish its disease connection. Bioinformatics tools help in validating targets by: Studying gene expression data Analyzing protein-protein interactions Predicting the effect of target inhibition The process confirms that a particular molecule target will achieve the expected treatment results.

Pharmacogenomics and Personalized Medicine Bioinformatics serves as the fundamental component which enables pharmacogenomics research to examine how genetic differences influence medication efficacy. The system helps to: Predict patient-specific responses to drugs Decrease the occurrence of harmful drug reactions Create customized treatment approaches This method enhances both medication safety and treatment success rates.

Applications of Machine Learning in Bioinformatics

Sequence analysis Scientists use machine learning to study the sequences of DNA and RNA and proteins. The ML models assist researchers in Detecting genes Identifying promoter regions and splice junctions Classifying proteins Finding preserved genomic sequences The techniques deliver better accuracy results than existing rule-based approaches.

Gene Expression Analysis Microarray and RNA-Seq experiments generate extensive and intricate gene expression data. Machine learning enables scientists to: Detect genes that show differential expression between different conditions Group genes that exhibit identical expression patterns Determine whether samples belong to the healthy or diseased category This application achieves particular significance for cancer research and disease diagnosis.

Disease Diagnosis and Classification

Machine learning functions as a vital component which enables medical bioinformatics to perform its task of analyzing biological data and clinical data. The system provides assistance for:

- The detection of diseases at their initial stage
- The process of classifying different types of cancer
- The process of discovering disease-specific biomarkers

Medical professionals use machine learning systems to achieve precise diagnostic results within a short time frame.

Advantages of Machine Learning in Bioinformatics

[illegible]

learning boosts bioinformatics research by making research processes more efficient and dependable.

Future Scope The future of machine learning in bioinformatics shows great potential for development. The technology will gain new uses through its combination with advanced technological systems. The upcoming developments will include: Deep learning for complex biological systems Integration of multi-omics data AI-based personalized healthcare Faster and more accurate drug discovery Improved disease prediction models Machine learning will continuously transform bioinformatics and biomedical research.

References

1. National Center for Biotechnology Information (NCBI)
2. UniProt Protein Database
3. Protein Data Bank (PDB)
4. Review articles on machine learning in bioinformatics
5. Bioinformatics and machine learning textbooks